

DATA ANALYSIS PROJECT

Global EV Market Analysis

A 2-page Power BI dashboard exploring EV sales, market share, and regional trends (2010–2025)

Source: IEA Global EV Data • Tools: Python (pandas), Power BI

Created by Rahul Jadhav

What This Project Does

An end-to-end analysis of the global electric vehicle transition — from raw IEA data to a cleaned dataset and a 2-page interactive Power BI dashboard.



Objective

Understand how EV sales, stock and market share have evolved globally, and let users explore by region and powertrain.



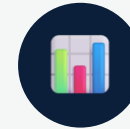
Dataset

IEA Global EV Data(2010-2025) - Spanning 71 countries , Covering EV sales, stock and market share metrics.



Cleaning (Python)

pandas used to remove duplicates, drop null regions, filter to key parameters, and export an analysis-ready CSV .



Visualization

2-page Power BI dashboard: an Executive summary page, plus an Explore page with Region and Powertrain slicers.

Data Cleaning Process

Performed in Python (pandas) via Jupyter Notebook, before loading into Power BI

01

Load & Inspect

Read the raw IEA Excel file; review shape, columns, dtypes and null counts.

02

Remove Duplicates

Identify and drop fully duplicated rows to avoid double-counting values.

03

Handle Missing Data

Drop records with a missing region_country, since they can't be attributed.

04

Filter Key Metrics

Keep only the 5 core parameters (sales, stock, sales share, stock share, charging points) out of 19 — cutting 48,282 rows down to the relevant subset.

05

Export Clean CSV

Save the analysis-ready dataset — for Power BI and further analysis.

A 2-Page Interactive Dashboard

Page 1 gives a fixed executive summary; Page 2 lets users explore by region and powertrain

PAGE 1

Executive Dashboard

- 1 slicer: Year
- 3 KPI cards: Global EV Sales, Global EV Stock, EV Sales Share
- Global EV Sales Trend (2010–2025) — line chart
- Major EV Markets Comparison: China, USA, Germany — line chart
- Top 10 EV Markets — bar chart
- BEV vs PHEV Sales Distribution — donut chart

PAGE 2

Explore by Region & Powertrain

- 2 slicers: Region/Country, Powertrain
- EV Sales Trend — line chart, responds to both slicers
- Powertrain Sales Distribution — stacked column by year
- China vs USA vs India — line chart (growth-market view)
- EV Stock Trend — line chart
- Selected Market Data — detail table

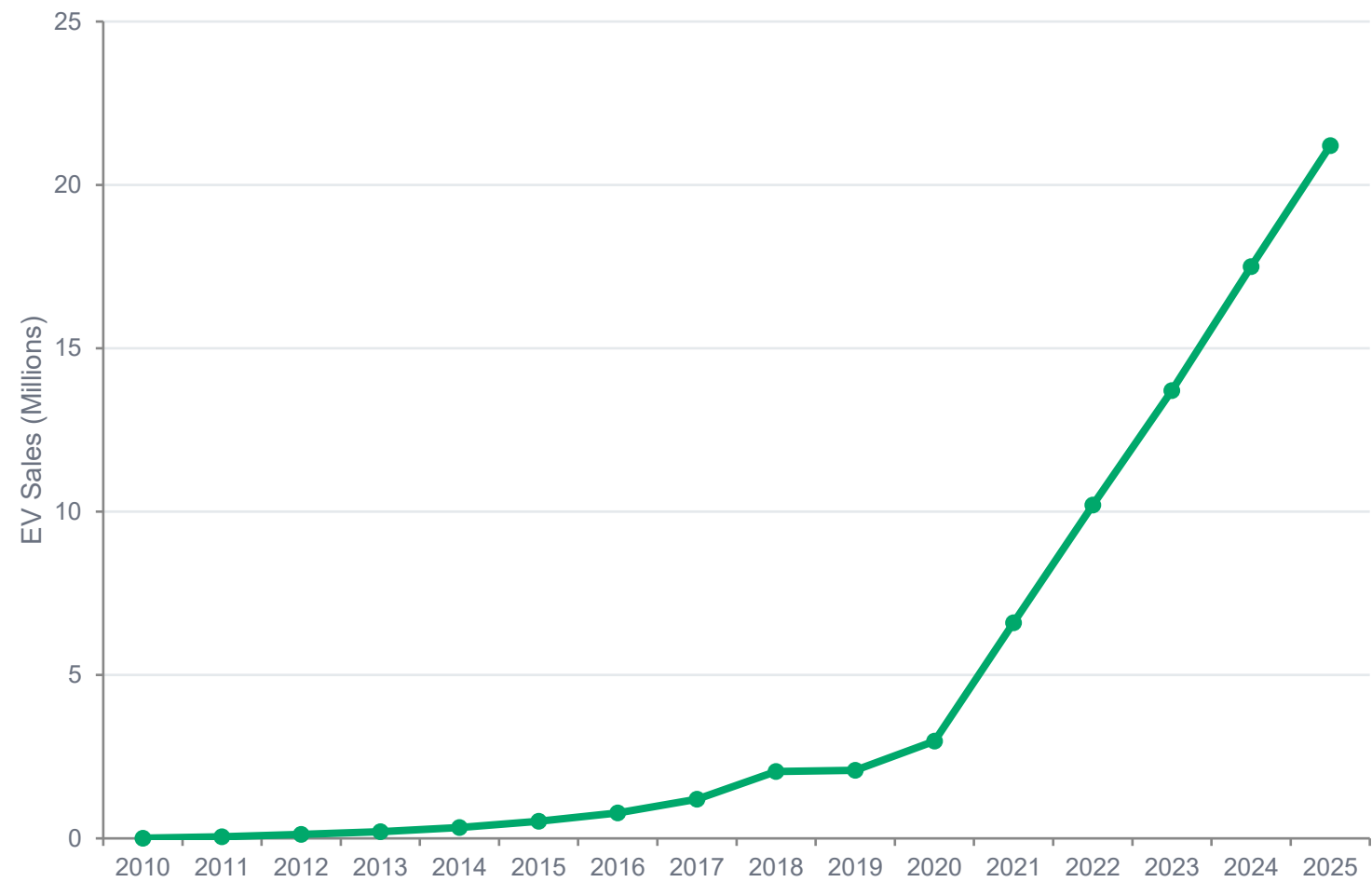
PAGE 1

Executive Dashboard

A fixed, high-level view — every chart shows the true global picture, filterable only by Year.

SALES TREND

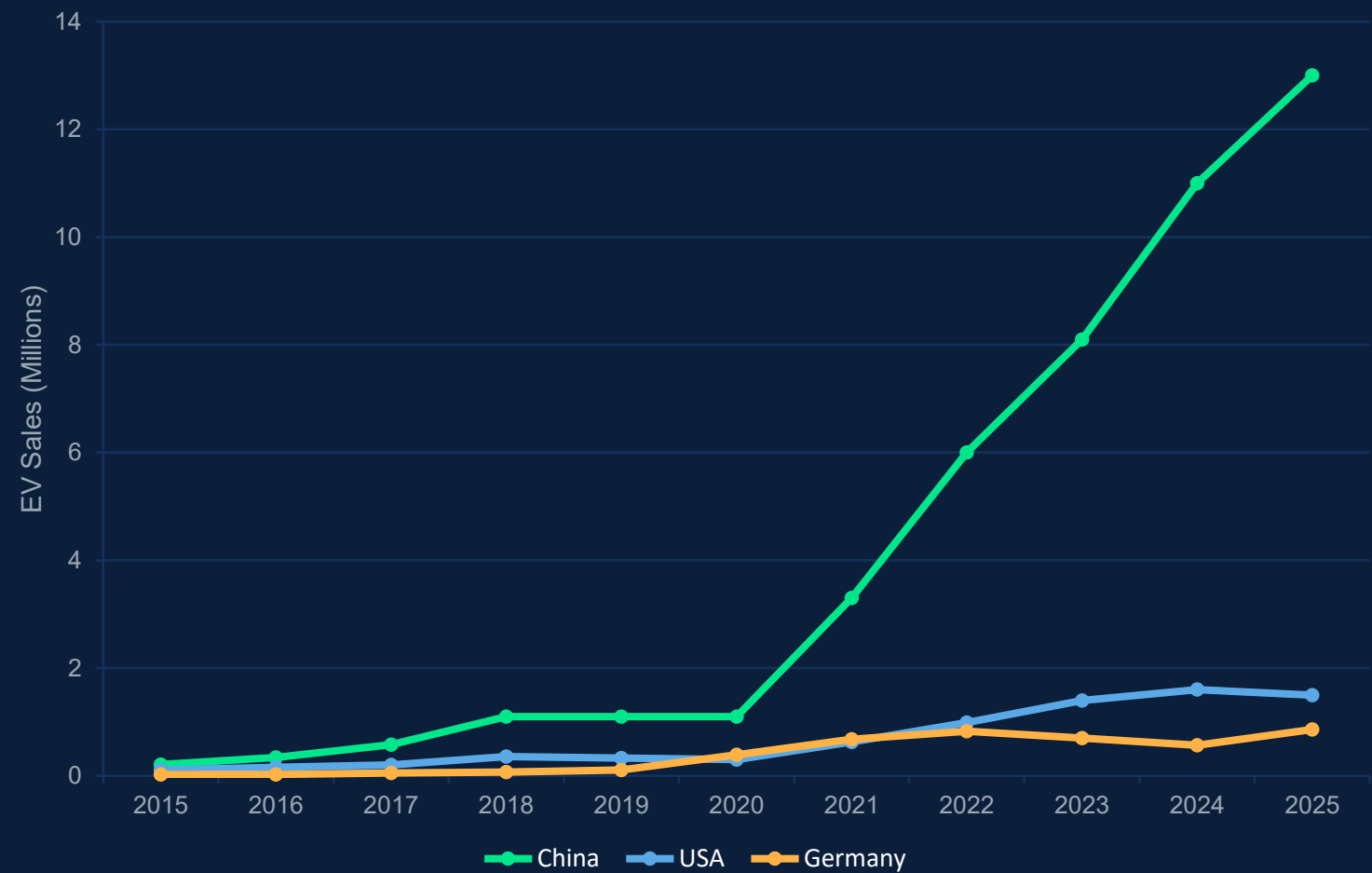
Global EV Sales Trend (2010–2025)



KEY INSIGHTS

- Sales grew from ~7,450 vehicles in 2010 to 20 million in 2025.
- Growth accelerated sharply after 2020, driven by incentives, falling battery costs and demand.
- 2021–2025 was the strongest growth period — the mass-adoption phase.
- World, Cars, EV powertrain — the fixed global baseline used across Page 1.

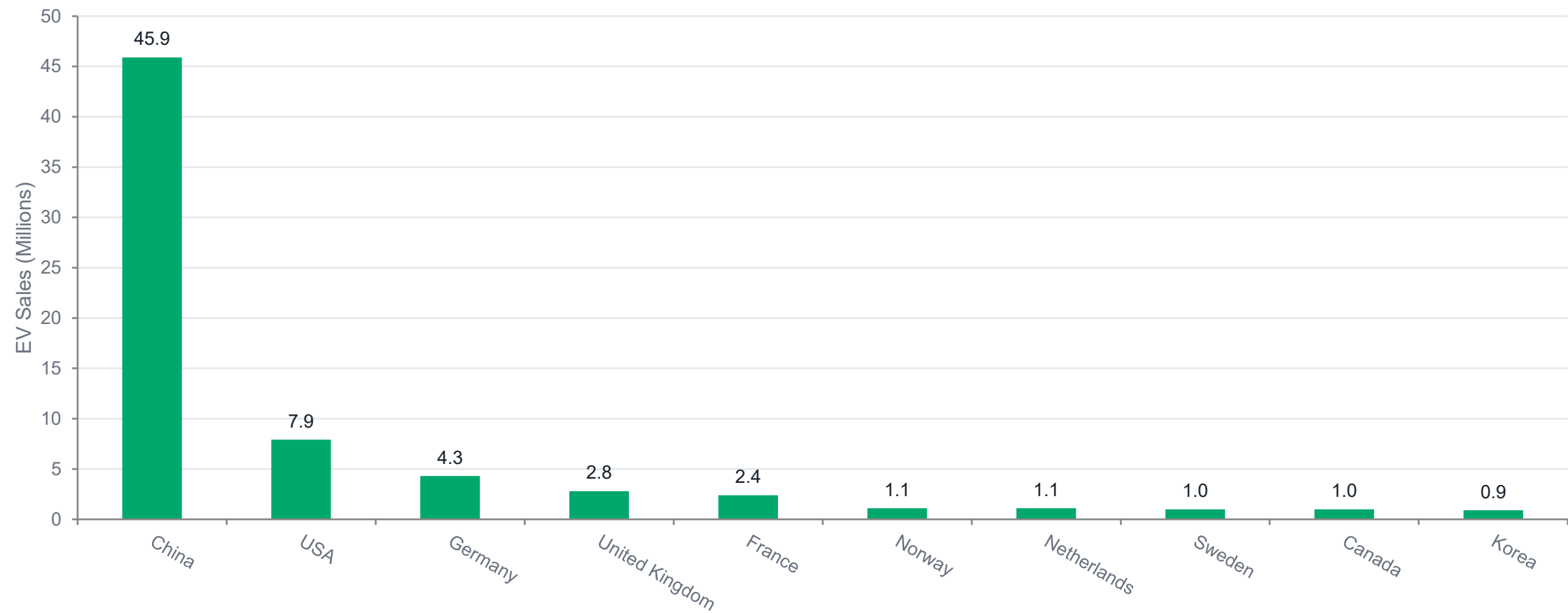
Major EV Markets Comparison: China, USA & Germany



KEY INSIGHTS

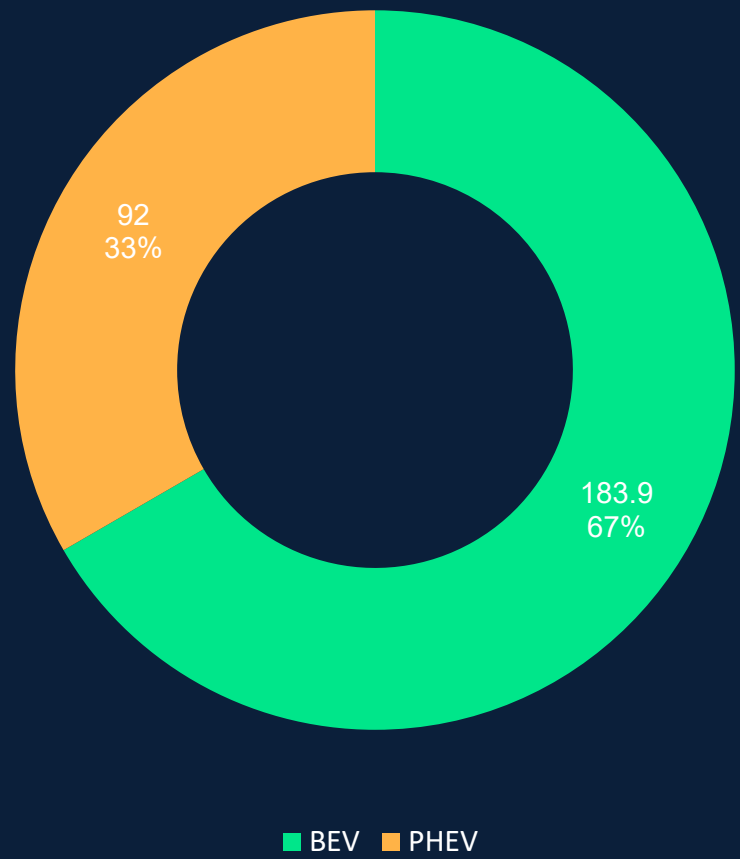
- These are the true top 3 markets by 2025 sales volume — not just the largest, most-established ones.
- China dominates by a wide margin: 13.0M in 2025, vs. USA's 1.5M and Germany's 0.86M.
- USA shows steady, consistent growth across the decade.
- Germany dipped in 2023–2024 (subsidy changes) before recovering in 2025.

Top 10 EV Markets (All-Time, Cars, EV)



Cumulative 2010–2025 sales, Cars & EV powertrain only (BEV+PHEV combined, no double-counting). China's cumulative total is ~5.8x the USA's.

BEV vs. PHEV Sales Distribution



KEY INSIGHTS

- Across all years combined, BEVs make up 66.7% of car sales vs. 33.3% for PHEVs.
- BEV sales (183.9M) outweigh PHEV sales (92.0M) roughly 2 to 1.
- This matches the yearly trend: BEV adoption has consistently outpaced PHEV.
- All-time, World, Cars only — the fixed global summary for Page 1.

SUMMARY

Page 1 — Dashboard KPIs

Snapshot shown: Year = 2025, World, Cars, EV powertrain — the same scope used across every Page 1 chart.

20.84M

Global EV Sales — World, 2025

75M

Global EV Stock — World, 2025

25%

EV Sales Share — World, 2025

WHY ONLY A YEAR SLICER ON THIS PAGE

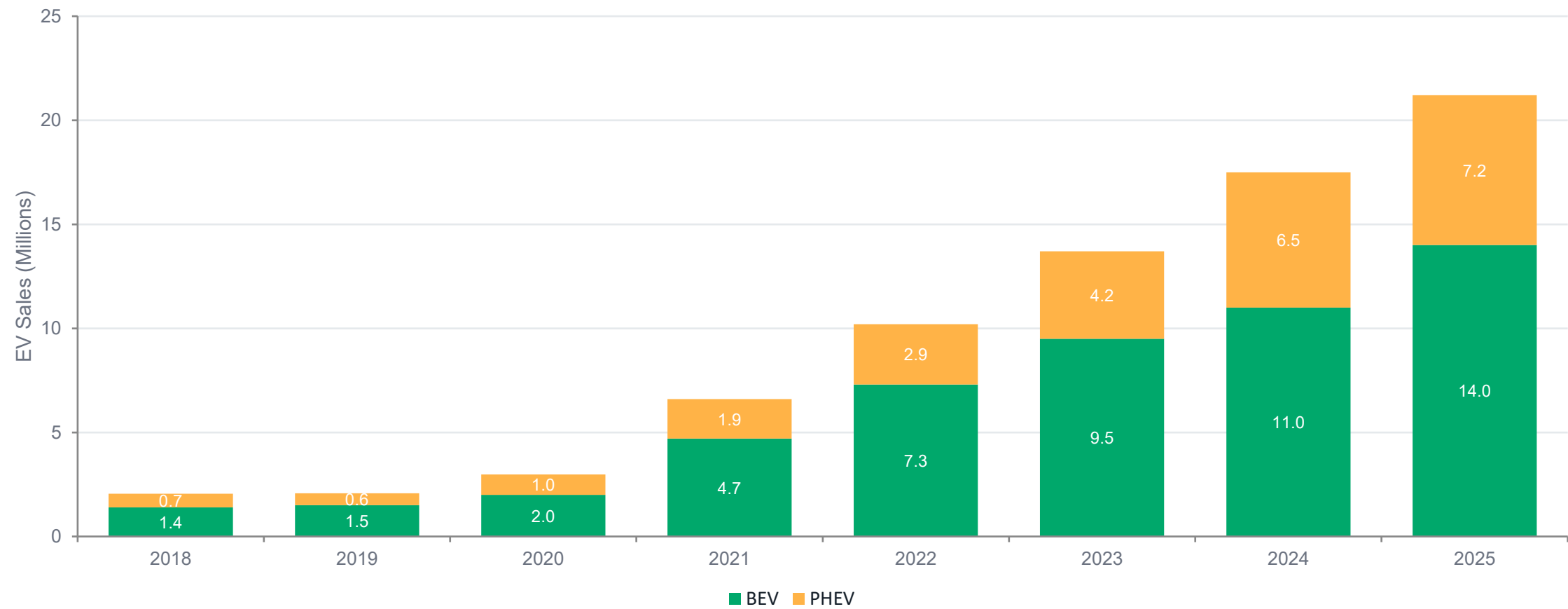
- Every visual here is titled "Global" or shows a fixed comparison — Region and Powertrain slicers would conflict with that promise.
- Region and Powertrain exploration now lives entirely on Page 2, where every chart is built to respond to them safely.

PAGE 2

Explore by Region & Powertrain

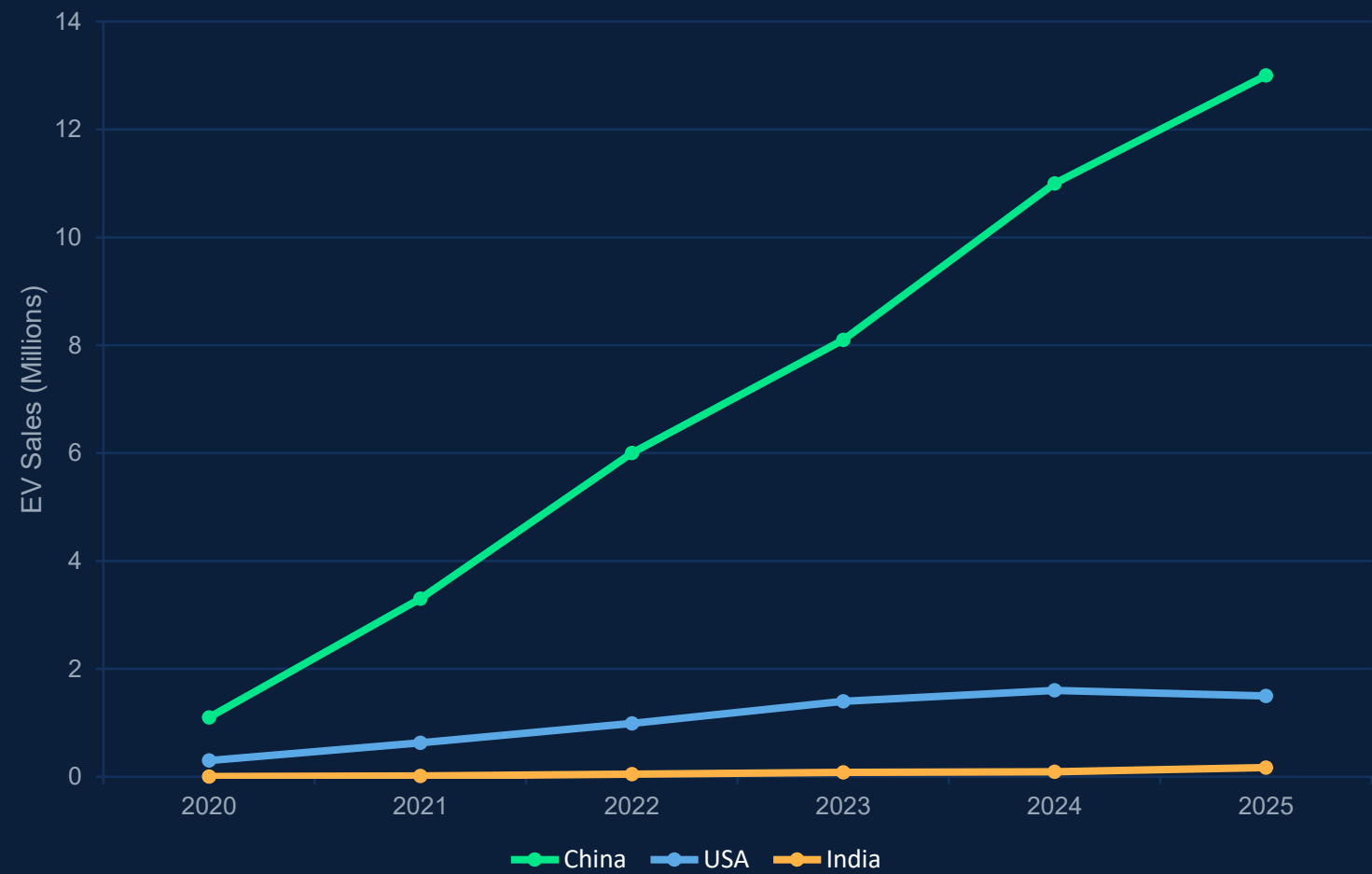
Two slicers — Region/Country and Powertrain — let users drill into the data beyond the fixed global view.

Powertrain Sales Distribution



World, Cars only. Unlike the Page 1 donut (all-time totals), this shows the widening BEV/PHEV gap year by year — filterable by Region and Powertrain slicers.

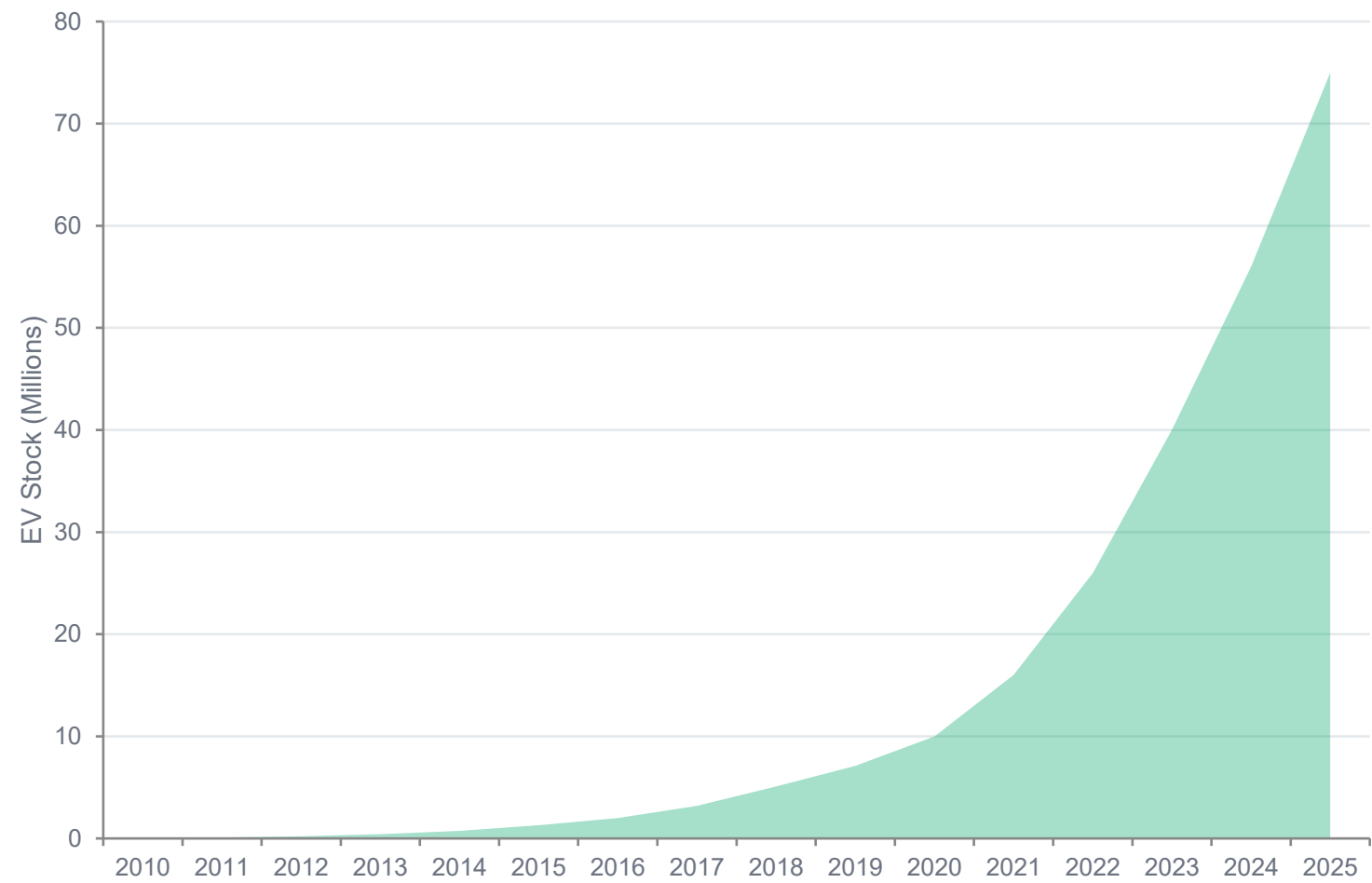
China vs. USA vs. India (Historical EV Sales)



KEY INSIGHTS

- Kept intentionally as a growth-market story, not a ranking — India is not currently a top-3 market by volume (Germany and the UK both outsell it).
- India's 2025 sales (170K) are still far behind USA (1.5M) and China (13.0M).
- But India's YoY growth rate is now accelerating faster than USA's, which is the point of this view.
- For an accurate ranking, see "Top 10 EV Markets" on Page 1.

EV Stock Trend



KEY INSIGHTS

- Global EV stock rose from ~39,000 vehicles in 2010 to 75 million in 2025.
- That's nearly a 1,900x increase in the number of EVs on the road in 15 years.
- The steepest acceleration happened after 2020, mirroring the sales trend.
- World, Cars, EV powertrain — new to Page 2, this trend wasn't charted on Page 1 (only shown there as a single KPI number).

Thank You

Global EV Market Analysis • Python (pandas) + Power BI • Data: IEA Global EV Data

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